

## J6-A, Netherlands

WMO: 062110

Lat: 53.824N

Lon: 2.945E

Elev: 34

StdP: 100.92

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 0.2        | 1.3 | -5.9                          | 2.3 | 1.2  | -4.6 | 2.6 | 2.1  | 19.3                  | 7.5  | 18.3 | 7.4  | 10.6                  | 120  | 0.853 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 2.0                    | 20.1            | 17.9 | 19.1 | 17.2 | 18.4 | 16.6 | 18.2                | 19.5 | 17.5 | 18.7 | 16.9 | 18.1 | 7.7                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.6                            | 12.6 | 19.1 | 16.9 | 12.1 | 18.5 | 16.3 | 11.6 | 17.9 | 51.5          | 19.6 | 49.5 | 18.8 | 47.6 | 18.2 | 21.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 18.2              | 16.3 | 14.8 | DB                         | 0.4  | 21.5               | 0.9 | 2.0                                                | -0.2 | 22.9       | -0.8 | 24.1       | -1.3 | 25.2       | -1.9 | 26.6 |
|                   |      |      | WB                         | -1.5 | 18.9               | 2.0 | 1.3                                                | -2.9 | 19.8       | -4.1 | 20.6       | -5.3 | 21.4       | -6.8 | 22.4 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 10.6        | 5.8   | 5.1   | 5.9   | 7.7   | 10.3  | 13.1  | 15.9  | 16.6  | 15.4  | 13.2  | 10.0  | 7.5   | (6)  |
|                                                                        | DBStd       | 4.42        | 1.98  | 2.16  | 1.85  | 1.81  | 1.93  | 1.57  | 1.69  | 1.40  | 1.64  | 1.65  | 2.15  | 2.04  | (7)  |
|                                                                        | HDD10.0     | 589         | 130   | 137   | 127   | 72    | 19    | 0     | 0     | 0     | 0     | 1     | 26    | 78    | (8)  |
|                                                                        | HDD18.3     | 2831        | 388   | 370   | 385   | 319   | 248   | 156   | 78    | 55    | 88    | 159   | 250   | 334   | (9)  |
|                                                                        | CDD10.0     | 807         | 0     | 0     | 0     | 3     | 30    | 94    | 183   | 206   | 163   | 101   | 26    | 2     | (10) |
|                                                                        | CDD18.3     | 7           | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 3     | 2     | 0     | 0     | 0     | (11) |
|                                                                        | CDH23.3     | 1           | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | (12) |
|                                                                        | CDH26.7     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 7.9         | 9.2   | 9.0   | 8.3   | 7.4   | 7.3   | 6.4   | 6.4   | 6.8   | 7.3   | 8.8   | 9.0   | 9.1   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 975         | 94    | 72    | 66    | 46    | 52    | 63    | 74    | 93    | 95    | 104   | 108   | 108   | (15) |
|                                                                        | PrecMax     | 1262        | 165   | 146   | 139   | 99    | 88    | 135   | 172   | 209   | 192   | 191   | 192   | 182   | (16) |
|                                                                        | PrecMin     | 724         | 12    | 23    | 10    | 3     | 24    | 19    | 13    | 20    | 19    | 37    | 18    | 50    | (17) |
|                                                                        | PrecStd     | 132         | 39    | 31    | 31    | 24    | 19    | 30    | 36    | 46    | 43    | 43    | 39    | 35    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.1  | 9.9   | 10.0  | 13.7  | 15.8  | 18.3  | 22.7  | 21.4  | 21.6  | 17.8  | 15.0  | 11.7  | (19) |
|                                                                        |             | MCWB        | 8.9   | 8.1   | 8.3   | 10.2  | 12.8  | 16.3  | 19.4  | 18.9  | 19.6  | 15.8  | 13.4  | 10.3  | (20) |
|                                                                        | 2%          | DB          | 9.4   | 8.9   | 9.1   | 11.9  | 14.6  | 16.8  | 20.1  | 19.9  | 19.2  | 17.0  | 14.3  | 11.2  | (21) |
|                                                                        |             | MCWB        | 8.2   | 7.5   | 7.8   | 9.5   | 12.4  | 15.2  | 18.1  | 17.4  | 17.6  | 15.3  | 12.9  | 10.0  | (22) |
|                                                                        | 5%          | DB          | 8.8   | 8.2   | 8.5   | 10.7  | 13.7  | 15.9  | 19.1  | 19.1  | 18.4  | 16.2  | 13.6  | 10.7  | (23) |
|                                                                        |             | MCWB        | 7.5   | 7.0   | 7.2   | 9.1   | 12.1  | 14.5  | 17.2  | 17.0  | 16.7  | 14.6  | 12.2  | 9.6   | (24) |
|                                                                        | 10%         | DB          | 8.3   | 7.7   | 8.0   | 9.9   | 12.9  | 15.1  | 18.1  | 18.5  | 17.7  | 15.4  | 12.9  | 10.1  | (25) |
|                                                                        |             | MCWB        | 7.1   | 6.6   | 6.7   | 8.7   | 11.6  | 13.7  | 16.3  | 16.6  | 16.0  | 13.8  | 11.4  | 9.0   | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.3   | 8.6   | 8.8   | 10.6  | 13.7  | 16.5  | 19.6  | 19.2  | 18.5  | 16.2  | 14.0  | 10.9  | (27) |
|                                                                        |             | MCDB        | 9.8   | 9.3   | 9.5   | 12.4  | 14.7  | 17.9  | 21.8  | 21.1  | 19.2  | 17.4  | 14.7  | 11.3  | (28) |
|                                                                        | 2%          | WB          | 8.5   | 7.8   | 8.1   | 9.9   | 12.9  | 15.6  | 18.3  | 18.1  | 17.7  | 15.5  | 13.2  | 10.4  | (29) |
|                                                                        |             | MCDB        | 9.1   | 8.5   | 8.9   | 11.2  | 14.0  | 16.6  | 19.9  | 19.3  | 18.6  | 16.7  | 14.0  | 11.0  | (30) |
|                                                                        | 5%          | WB          | 7.9   | 7.3   | 7.4   | 9.4   | 12.2  | 14.8  | 17.5  | 17.5  | 17.0  | 14.8  | 12.5  | 9.8   | (31) |
|                                                                        |             | MCDB        | 8.6   | 8.0   | 8.2   | 10.6  | 13.3  | 15.8  | 18.8  | 18.7  | 18.1  | 15.9  | 13.4  | 10.6  | (32) |
|                                                                        | 10%         | WB          | 7.3   | 6.8   | 7.0   | 8.8   | 11.7  | 14.0  | 16.6  | 16.9  | 16.3  | 14.2  | 11.6  | 9.1   | (33) |
|                                                                        |             | MCDB        | 8.1   | 7.5   | 7.7   | 9.9   | 12.8  | 14.9  | 17.7  | 18.2  | 17.4  | 15.3  | 12.7  | 10.0  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 2.1   | 1.9   | 1.9   | 2.1   | 2.0   | 1.9   | 2.0   | 2.0   | 2.0   | 2.1   | 2.2   | 2.1   | (35) |
|                                                                        |             | MCDBR       | 2.3   | 2.3   | 2.3   | 2.9   | 2.9   | 2.6   | 2.9   | 2.5   | 2.4   | 2.5   | 2.7   | 2.2   | (36) |
|                                                                        |             | MCWBR       | 2.7   | 2.1   | 2.0   | 2.0   | 2.2   | 2.3   | 2.0   | 2.4   | 2.4   | 2.7   | 3.1   | 2.8   | (37) |
|                                                                        | 5% WB       | MCDBR       | 2.2   | 2.3   | 2.3   | 2.8   | 2.7   | 2.6   | 2.7   | 2.4   | 2.3   | 2.3   | 2.7   | 2.3   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.332 | 0.347 | 0.375 | 0.401 | 0.394 | 0.400 | 0.404 | 0.394 | 0.382 | 0.360 | 0.335 | 0.319 | (40) |
|                                                                        |             | taud        | 2.306 | 2.356 | 2.313 | 2.276 | 2.378 | 2.373 | 2.371 | 2.409 | 2.431 | 2.424 | 2.356 | 2.296 | (41) |
|                                                                        |             | Ebn at Noon | 616   | 742   | 799   | 825   | 854   | 851   | 839   | 826   | 787   | 719   | 609   | 550   | (42) |
|                                                                        | Edh at Noon | Edh at Noon | 60    | 78    | 100   | 118   | 112   | 114   | 112   | 103   | 89    | 73    | 57    | 50    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.61  | 1.33  | 2.67  | 4.40  | 5.58  | 5.74  | 5.58  | 4.68  | 3.18  | 1.70  | 0.74  | 0.46  | (44) |
|                                                                        | RadStd      | RadStd      | 0.06  | 0.21  | 0.21  | 0.42  | 0.37  | 0.35  | 0.56  | 0.32  | 0.27  | 0.12  | 0.07  | 0.04  | (45) |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (28 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -110    | +52     | N/A     |

Nomenclature: See separate page